

INPUTS – pre-tournament feeds

- baseline Elo ratings R_0 (per team)
- injury & host-advantage adjustments
- structure S : groups, fixtures, bracket seeding
- Monte-Carlo settings: N runs, random seed
- de-vigged bookmaker 1X2 odds $\rightarrow$ group λ
- prediction-market (Polymarket) champion odds
- proxy-xG coefficients (fit on 2018+2022)

1. **Initialise** frozen $\leftarrow R_0$ $L \leftarrow \text{LearningTrack}(R_0, \text{drift} = 0)$ results $\leftarrow \emptyset$

2. $t=0$ baseline snapshot: simulate both tracks $\rightarrow$ champion + per-stage distribution

LOOP $\Rightarrow$ for each game g , in the order the tournament plays them

3. BEFORE kickoff – gather & feed (per match)

- confirmed starting XI (both teams)
- late injury / suspension / illness news
- rotation & key-player availability
- formation / tactical notes
- venue, altitude, weather, pitch, referee
- rest days, travel, fixture congestion
- strengths learned so far (L , accumulated drift)
- live market champion-odds snapshot

4. simulate both tracks at the current state $\rightarrow$ record *before* snapshot (+ KL vs. previous)

... game g is played ...

5. AFTER full-time – gather & feed (per match)

- final result / scoreline (90'; ET/pens winner)
- goal & card timeline (minute by minute)
- shots; shots on / off target; blocked
- possession %; passes; pass accuracy
- all combined $\rightarrow \lambda_{\text{obs}}$ (observed expected goals, per team)
- corners, fouls, offsides
- yellow / red cards
- big chances; xG: proxy (shots-on-target), or free real xG (FBref / FotMob)
- proxy-vs-real-xG agreement (diagnostic)

6. condition: results $\leftarrow \text{results} \cup \{g\}$ (knockout: record the advancing team)

7. **LEARN** $\lambda_{\text{exp}} \leftarrow \text{ratings}$; surprise $\delta \leftarrow (\lambda_{\text{obs}} - \lambda_{\text{exp}})_{\text{for}} - (\lambda_{\text{obs}} - \lambda_{\text{exp}})_{\text{against}}$;
 $L.\text{apply_match}$: bounded regularised-Elo drift (zero-sum, shrunk to R_0 , swept knob k)

8a. simulate **FROZEN** (R_0) $\rightarrow$ champ. dist

8b. simulate **LEARNING** (L) $\rightarrow$ champ. dist

9. record *after* snapshot (both tracks) + KL vs. previous + info_source tag (result / performance)

more
games?

yes, next g

OUTPUTS

- **trajectory** of before/after snapshots (both tracks)
- champion + per-stage probabilities over time
- per-team strength **drift** over time
- swept-knob (learning-rate) sensitivity curve
- KL information per game (what mattered)
- pivotality / outcome-surprise per game
- **A-vs-B** gap: did learning beat conditioning?
- model-vs-market divergence over the run